

Graphing Rational Functions

ANSWER KEY



Reading graphs

- 1 The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.
Vertical asymptote $x = 3$; horizontal asymptote $y = 1$; intercepts $(-2, 0)$ and $(0, -\frac{2}{3})$.
- 2 The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.
 - a Describe the transformation. Shift right 2 units and up 3 units.
 - b State the asymptotes of g . Vertical $x = 2$; horizontal $y = 3$.
 - c State the domain and range. Domain $x \neq 2$; range $y \neq 3$.

Sign analysis

- 3 Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.
Testing each interval: $(-\infty, -2)$ negative; $(-2, 1)$ positive; $(1, 4)$ negative; $(4, \infty)$ positive. So $f(x) > 0$ on $(-2, 1) \cup (4, \infty)$.
- 4 Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.
Critical values $x = -3$ (zero) and $x = 2$ (undefined). The quotient is negative between them: $[-3, 2)$ — including -3 , excluding 2 .
- 5 Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.
Vertical asymptotes $x = 1$ and $x = -1$; horizontal asymptote $y = 0$. Since $h(-x) = -h(x)$, the function is odd — symmetric about the origin.